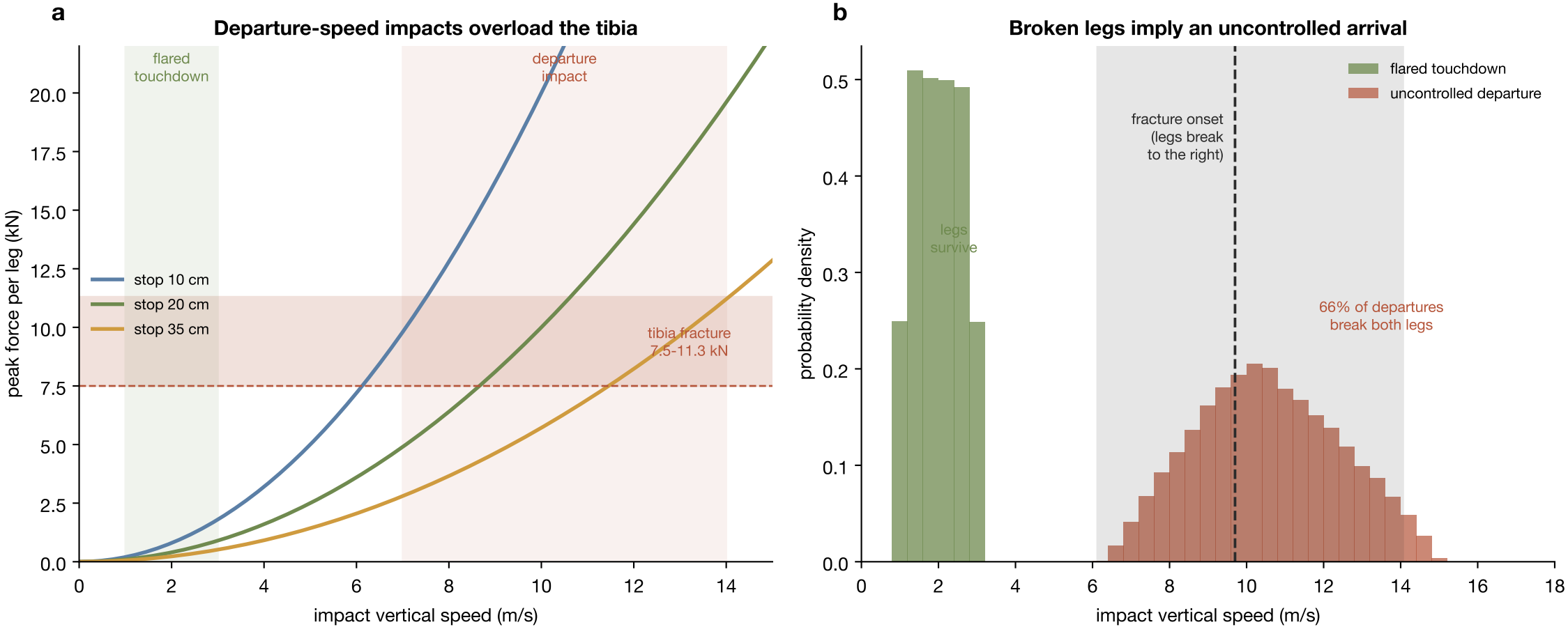
a
Departure-speed impacts overload the tibia
flared touchdown
departure impact
stop 10 cm
stop 20 cm
stop 35 cm
tibia fracture 7.5-11.3 kN
peak force per leg (kN)
impact vertical speed (m/s)
b
Broken legs imply an uncontrolled arrival
flared touchdown
uncontrolled departure
fracture onset (legs break to the right)
legs survive
66% of departures break both legs
probability density
impact vertical speed (m/s)